

Lecture 3

17/11/16

Chapter 1: Sequence Alignment Algorithms

1.1: Edit graph & alignments

1.2: Scoring Alignments, scoring scheme

$X = x_1, x_2, \dots, x_M$

$Y = y_1, y_2, \dots, y_N$

$x_i, y_i \in \Sigma = \{A, C, G, T\}$

- $\begin{cases} s(x_i, y_i) \\ s(x_i, -) \\ s(-, y_i) \end{cases}$

most basic is:

s	-	A	C	G	T
-	0	0	0	0	0
A	0	1	0	0	0
C	0	0	1	0	0
G	0	0	0	1	0
T	0	0	0	0	1

unit

eg.

- edit graph, $G_{x,y}$, is an edge labelled directed graph

$\hookrightarrow G = (V, E)$, where V = vertex set and E = edge set

$\hookrightarrow$ Vertices are given by $(M+1)(N+1)$, start numbering from 0

$\hookrightarrow$ Edges are fundamental units of the alignment. There are three types of edges:

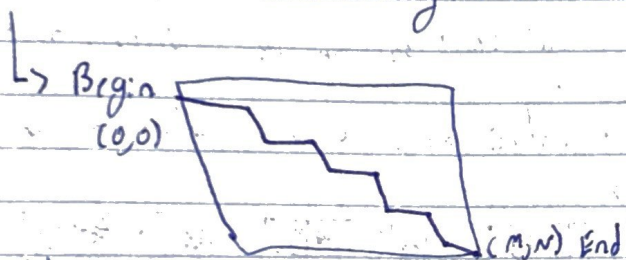
$\hookrightarrow$ Gap in y : $1 \leq i \leq M, 1 \leq j \leq N. (i-1, j) \xrightarrow{\begin{pmatrix} x_i \\ - \end{pmatrix} \text{ label of the edge}} (i, j)$

↳ Gap in X: $1 \leq i \leq M, 1 \leq j \leq N, (i, j-1) \xrightarrow{(x_i)} (i, j)$

↳ Alignment edge: $1 \leq i \leq M, 1 \leq j \leq N, (i-1, j-1) \xrightarrow{(y_i)} (i, j)$

↳ Only 3 types of edges coming to (i, j)

↳ There is a 1:1 correspondence between directed paths from $(0, 0)$ to (M, N) and alignments between X and Y



DAG: - Directed

- Acyclic

- Graph

- It is quadratic time to solve this and find the max alignment

Dynamic Programming:

- Denote $SC(i, j)$ the score of maximum path from start to (i, j)

↳ Because there are only three edges directed into vertex (i, j) it follows that any optimal path (maximum), P_{opt} , to (i, j) must use one of these edges:

1) P_{opt} first arrives at $(i-1, j)$ and it follows a gap in Y edge into (i, j)

2) P_{opt} first arrives at $(i, j-1)$ and then through a gap in X edge gets to (i, j)

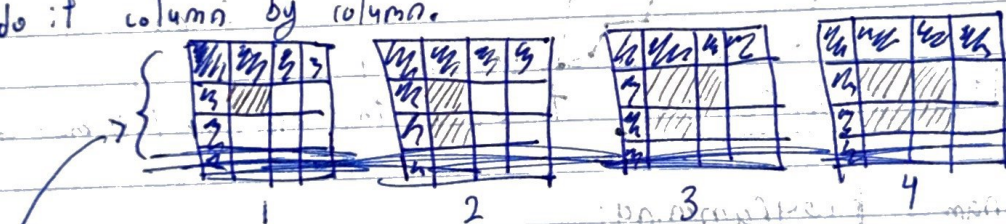
3) P_{opt} gets to $(i-1, j-1)$ and through an alignment edge gets to (i, j)

the optimal cost of getting to i, j

$$SC_i(j) = \text{MAX} \begin{cases} SC_{i-1}(j-1) + g(x_i, y_j) \\ SC_{i-1}(j) + g(x_i, -) \\ SC_i(j-1) + g(-, y_j) \end{cases}$$

- We need to compute the $SC_i(j)$ such that at any time we have already ~~completely~~ computed the values of S needed by the new node (at least one of the indices i and j smaller)

- there are several ways to solve such a computation. We will do it column by column.



Dynamic programming discipline

- say this is optimal.  How can we prove that this subsequence alignment is optimal? ~~If we were to~~
- If it was not optimal, then we would be able to find a shorter path in the total optimal path, but since this subsequence must be optimal.

The Global Alignment Algorithm:

$S[0,0] = 0$
for $j = 1$ to N do
 $S[0,j] = S[0,j-1] +$

Because any optimality is divided through optimal subpaths

The Global Alignment Algorithm:

$$S[0,0] = 0$$

For $j=1$ to N Do

$$S[0,j] = S[0,j-1] + g(-, y_j)$$

For $i=1$ to M Do

$$S[i,0] = S[i-1,0] + g(x_i, -)$$

For $j=1$ to N Do

$$S[i,j] = \max \begin{cases} S[i-1,j-1] + g(x_i, y_j) \\ S[i-1,j] + g(x_i, -) \\ S[i,j-1] + g(-, y_j) \end{cases}$$

- this algorithm is $O(MN)$, and if $M=N$, $O(N^2)$. So it's a quadratic algorithm.

conservation windows

W W Y I R

B A B A

W F Y V R

A A A C

W Y Y V R

A A C C

W Y Y T R

A A B A

W F Y R

A A C C

W Y Y V R

A A B C

There are 24 (4!) amino acids

B A B A
 A A A C
 A A C C
 A A B A
 A A C C
 A A B C

- there are 24 amino acids observed: - 14 are A
 (4+6) - 4 are B
 - 6 are C

- the observed proportions are: $-\frac{14}{24} =$
 $-\frac{4}{24} =$
 $-\frac{6}{24} =$

- Pairs of letters aligned to each other: $4 \binom{6}{2} = 4 \times \frac{6 \times 5}{2} = 60$
 ↳ there are 60 pairs aligned

aligned pair	frequency observed:
A to A	26/60
A to B	8/60
A to C	10/60
B to B	3/60
B to C	6/60
C to C	7/60

Need a random model for which to compute expectations

- Now we compare observed frequencies to expected:

↳ We will generate random sequences using the frequencies of amino acids ($A \frac{14}{24}, B \frac{4}{24}$) that is, we have chose at random sequences with the same frequency of amino acids given above, and make an alignment on them.

↳ in random alignment: A to A: $\frac{14}{24} \times \frac{14}{24}$ (independence)
 A to B: $2 \times \frac{14}{24} \times \frac{4}{24}$ ($\begin{smallmatrix} A \text{ to } B \\ B \text{ to } A \end{smallmatrix}$ order don't matter)
 etc..

↳ these fractions are used to calculate the estimated: $2 \log_2 \left(\frac{\text{observed}}{\text{expected}} \right)$
 likelihood ratios

Aligned Pair	Observed Frequency	Expected Frequency	$2 \log \left(\frac{\text{obs. frequency}}{\text{expected frequency}} \right)$
A to A	26/60	196/576	0.7
A to B	8/60	112/576	-1.09
A to C	10/60	168/576	-1.61
B to B	3/60	16/576	1.70
B to C	6/60	48/576	0.53
C to C	7/60	36/576	1.83

round up numbers to get scoring matrix

scoring scheme

	A	B	C
A	1	-1	-2
B	-1	2	1
C	-2	1	2